HEINZ NIXDORF INSTITUT
UNIVERSITÄT PADERBORN

Knowledge Graphs: Construction, Reasoning, and Retrieval-Augmented Generation

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DICE
Data Science Group
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Project Group - WiSe 2026
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Knowledge Graphs

Introduction

A Knowledge Graph (KG) represents entities as nodes and their relationships as edges.

- Knowledge Graphs Everywhere



- Many open knowledge graphs



- Powering AI



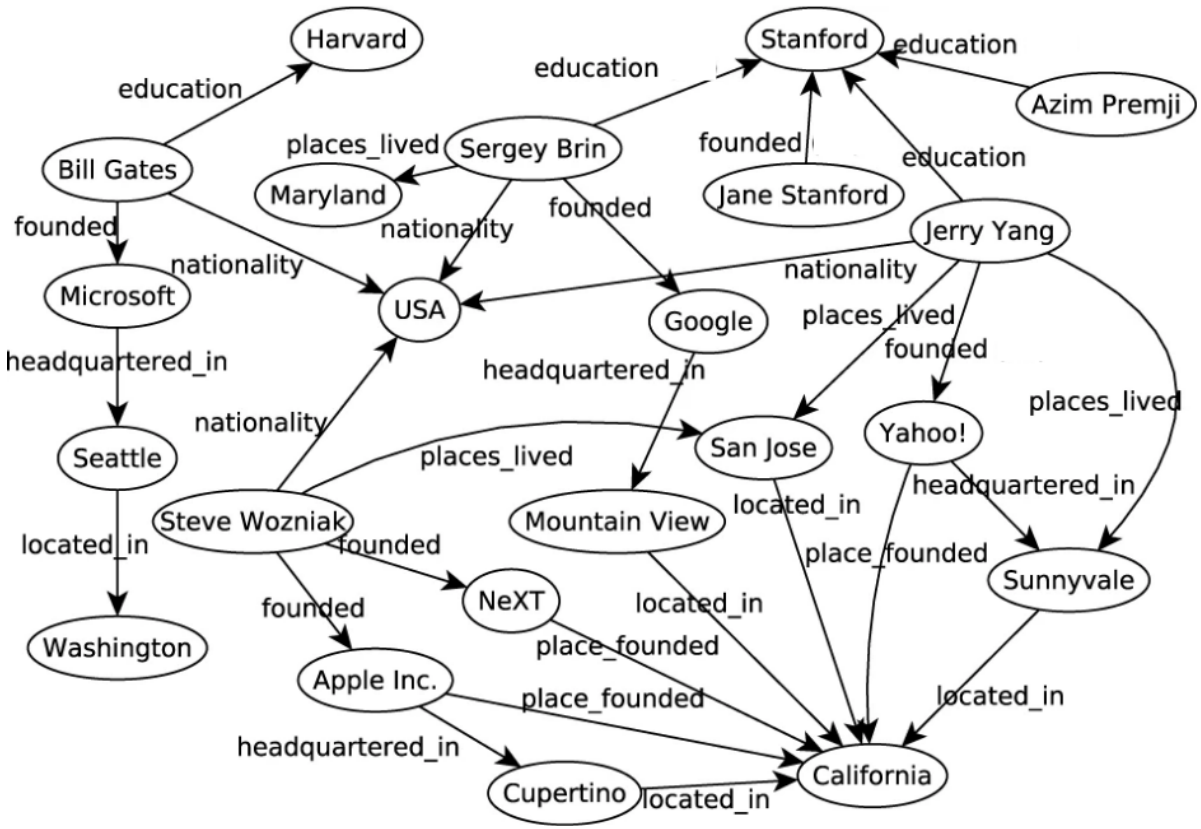
- Applied across diverse domains



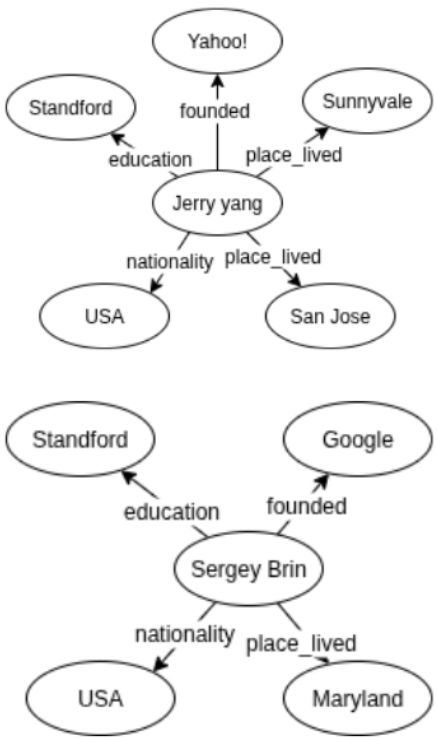
Knowledge Graphs

Structure and Representation

A knowledge Graph



Entity Description

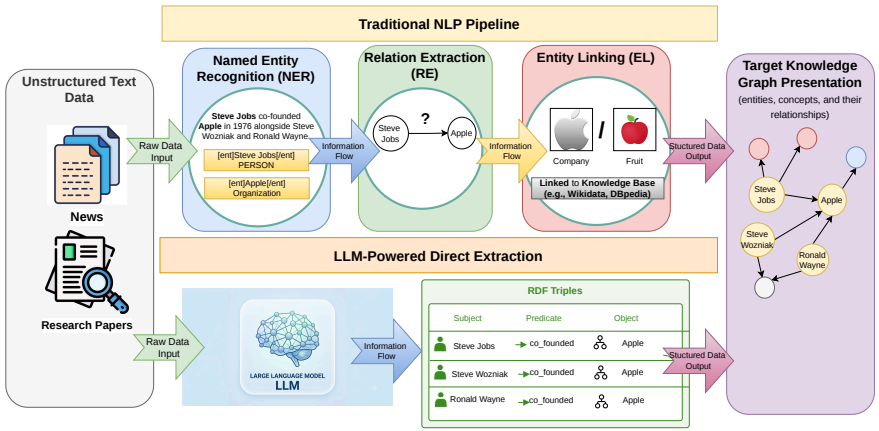


Knowledge Graphs: Construction

Why we need to construct the knowledge graph?

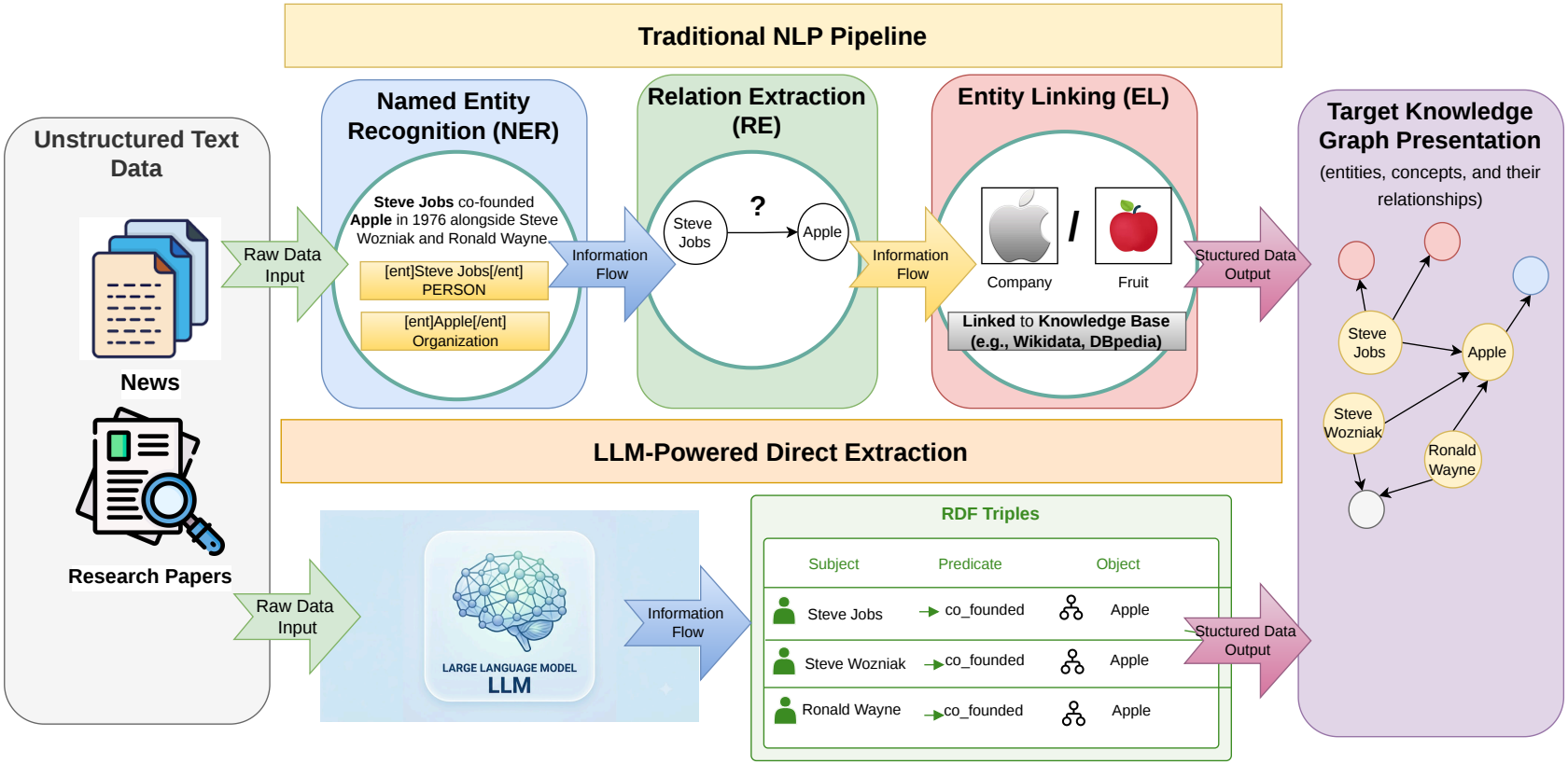
Motivation

- **Problem**: Vast amounts of valuable domain knowledge remain trapped in unstructured raw text.
- **Solution**: Extract structured RDF triples using NLP pipelines (NER, RE, EL) or direct LLMs.
- **Goal**: Build machine-readable, interconnected knowledge graphs from unstructured data sources.



Knowledge Graphs: Construction

Approaches

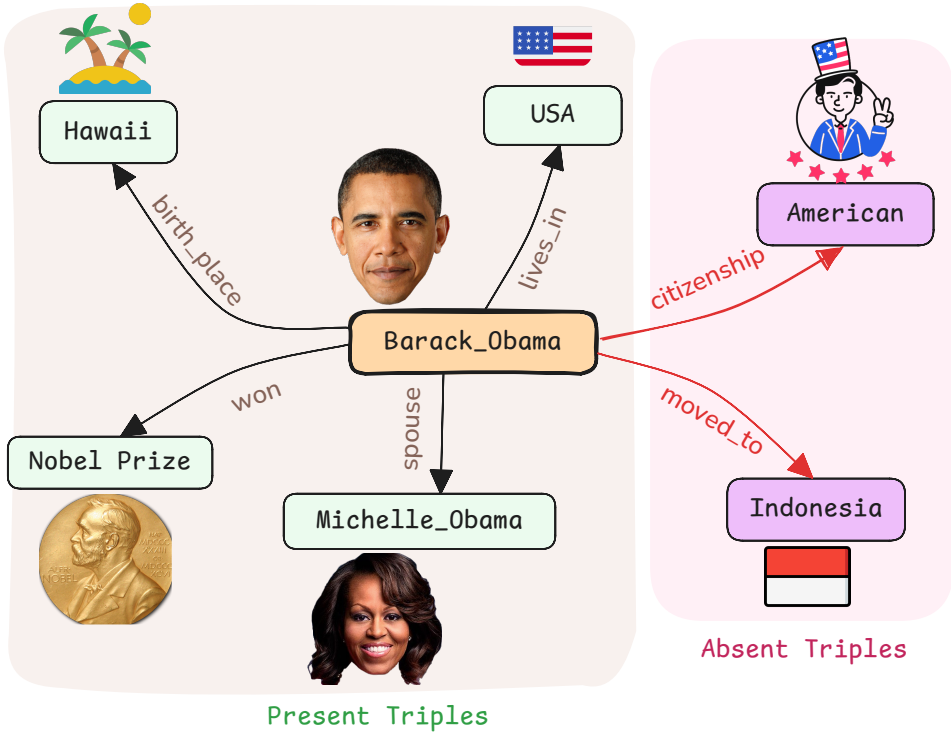


Tiwari, B., Oliviera, I., Firmansyah, A.F., Zahera, H.M., HOPFGARTNER, F., Ngomo, AC.N. (2026). Ontology-Aware Prompting for Knowledge Graph Construction from Text. Accepted at SEMANTICS 2026.

Knowledge Graphs: Reasoning

Motivation

- **Problem**: Incompleteness and missing links in Knowledge Graphs limit downstream tasks.
- **Solution**: Apply logical inference and machine learning (GNNs/Embeddings) for link prediction.
- **Goal**: Infer unstated facts, discover hidden patterns, and complete missing graph relationships.

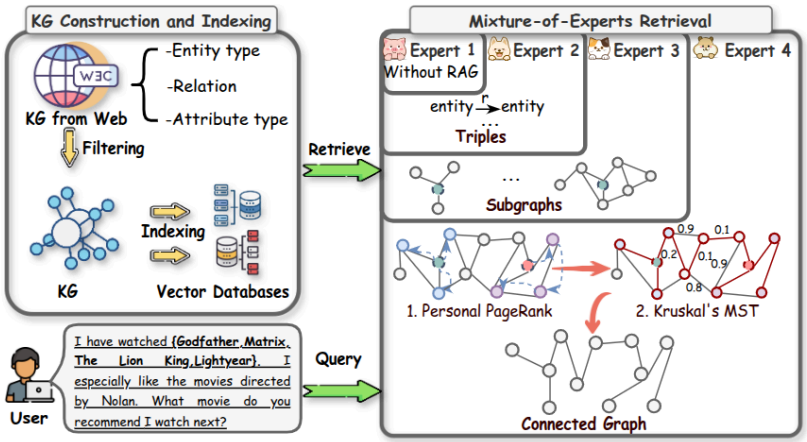


Firmansyah, A.F., Zahera, H.M., Sherif, M.A., Moussallem, D., Ngomo, AC.N. (2025). ANTS: Abstractive Entity Summarization in Knowledge Graphs. The Semantic Web – ESWC 2025. LNCS, vol. 15718, Springer, Cham.

KG Retrieval-Augmented Generation

Motivation

- **Problem**: LLMs suffer from hallucinations, lack domain-specific knowledge, and cannot cite explicit sources.
- **Solution**: Retrieve structured, factual subgraphs from Knowledge Graphs to ground LLM prompts in verifiable context.
- **Goal**: Generate accurate, explainable, and hallucination-free answers enriched with structured semantic knowledge.



Source: https://arxiv.org/pdf/2605.28175

Project Objectives

Addressing the core challenges of Knowledge Graph construction, reasoning, and RAG across the complete research lifecycle.

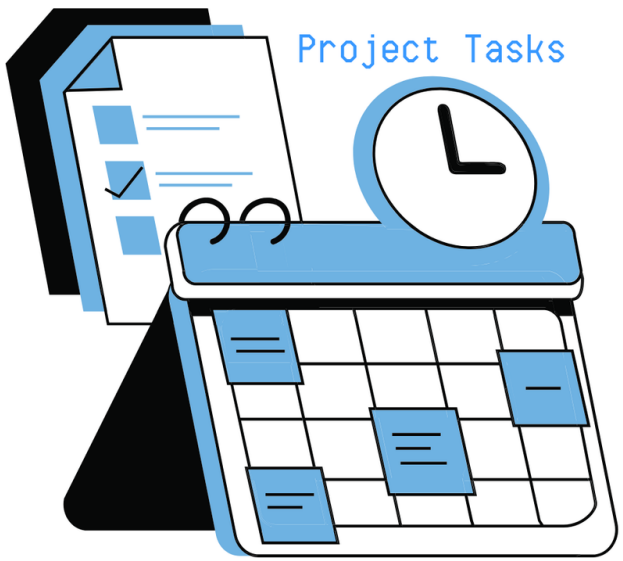
- **Enhance performance:** Improve extraction accuracy, logical inference capability, and factual precision in LLM generation.
- **Explore Advanced Techniques:** Leverage LLMs for automated RDF extraction, Deep Learning for KG reasoning, and KG-RAG architectures.
- **Discover Novel Approaches:** Innovate at the intersection of structured knowledge bases and Generative AI.



Project Tasks

What core tasks will the project group undertake across the research lifecycle?

- **Literature Study:** Survey state-of-the-art methods.
- **Baseline Reproduction:** Implement and test existing baselines on benchmark datasets.
- **Gap Analysis:** Identify limitations in current systems.
- **Method Development and Evaluation:** Design, implement, and evaluate improved pipelines for automated construction, reasoning, or KG-RAG.
- **Graph Visualization**

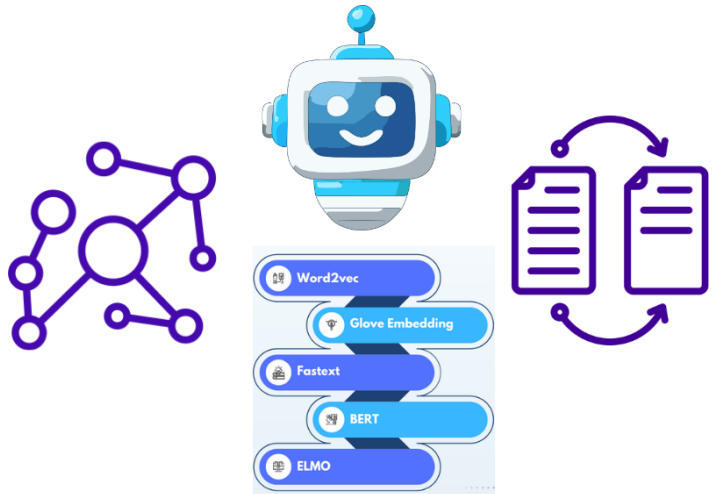


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Thank you!

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Have questions?

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